

Robert C. Bridges

(609) 647-9976 | Robert.Bridges609@gmail.com | linkedin.com/in/Robert-Bridges609 | robertcbridges.github.io/portfolio

PROFESSIONAL SUMMARY

Aerospace and mechanical engineer (M.Eng. Aerospace, B.S. Mechanical) with coursework in propulsion, combustion, and space structures and hands-on hardware experience across design, analysis, assembly, integration, and test. Authored SRR, SDR, and PDR documentation for an original spacecraft mission design and took a robotic bat from concept through fabrication and test. Proficient in CAD (Fusion 360, equivalent to Creo), FEA/CFD, and GD&T. Takes ownership of technical problems through to a solution and communicates clearly with technical and non-technical audiences.

EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Master of Engineering, Aerospace Engineering

May 2026

Bachelor of Science, Mechanical Engineering

Dec 2025

Relevant Coursework: Propulsion, Aircraft and Rockets; Combustion Processes; Plasmas for Propulsion; Space Structures; Spacecraft Technology and Systems; Introduction to Spaceflight Mechanics; Mechanical Metallurgy

TECHNICAL SKILLS

- **CAD & Analysis:** Fusion 360 (CAD equivalent to Creo), Ansys (FEA/CFD), GD&T, engineering drawing interpretation, mechanical design
- **Hardware:** design, analysis, assembly, integration, and test; hand and machine tools (Red Apron Machine Shop), Micro Vu, 3D printing (PLA, TPU, resin)
- **Aerospace & Materials:** propulsion and combustion fundamentals, spacecraft systems, lifecycle reviews (SRR, SDR, PDR), composites (honeycomb sandwich panels, carbon fiber layups), structural analysis
- **Documentation & Software:** BoM, technical documentation, design review packages, MS Office Suite, Adobe Creative Suite, Python, MATLAB, Java

ENGINEERING PROJECTS

Original Spacecraft Mission Design, F.O.C.A

Jan - May 2026

- Designed a spacecraft to intercept foreign objects in orbit, using prior flight missions as the architecture baseline.
- Authored SRR, SDR, and PDR documentation to validate the design and defined a full mission life timeline.

Robotic Bat Research, Master's Project

Aug 2025 - May 2026

- Designing and building a robotic bat, including its structure, circuitry, motors, and wing membrane, taking the hardware from concept through fabrication, assembly, integration, and test.
- Fabricate and iterate on structural components using PLA, TPU, and resin 3D printing.

Focus Glasses

Aug 2025 - Dec 2025

- Led market research and delivered a Product Design Specification with a 4-person team for glasses that keep users focused using invisible spectral flicker; completed FEA to confirm safety.
- Voted best new product in the class after presenting at a public exhibit.

Additional Projects

- **Van Allen Belt Electron Simulation** (Jan - May 2024): modeled the spacecraft radiation environment by simulating outer belt electron acceleration during a May 2024 solar storm (up to 2.4 MeV).
- **Spin Wheel, Senior Design** (Jan - May 2025): designed, prototyped, and filed a patent application for a novel computer scroll wheel.
- **Chainsaw Dissection** (Aug - Dec 2024): deconstructed a chainsaw to analyze its mechanical components and fluid dynamic properties.

PROFESSIONAL EXPERIENCE

Abbott Point of Care, Princeton, NJ

Project Engineering Intern

May - Aug 2022 & 2023

- Built and maintained a database of custom parts used across Abbott's medical devices, streamlining search and documentation for engineering teams (2023).
- Analyzed engineering bills of materials (BoM) and product structures for hundreds of components to support design and manufacturing decisions.
- Used precision measurement equipment, including Micro Vu, to inspect circuit boards; updated and calibrated TEAM, a company-wide database for tracking product errors (2022).
- Documented product issues and test results to strengthen troubleshooting and quality processes throughout the design cycle.

Princeton University, Facilities Department, Princeton, NJ

Engineering Intern

May - Aug 2025 & 2026

- Coordinated communication between project managers and on-site staff, producing daily progress reports to keep stakeholders aligned.
- Read and interpreted technical architectural plans and drafted a granite sign engraving template to support facility construction projects.

TEACHING & MENTORSHIP

- Teaching Assistant, Introduction to Spaceflight Mechanics & Introduction to Mechanical Design, Cornell MAE Department: graded assignments, led weekly labs, hosted office hours, and taught aerospace and mechanical design (2025 - Present).
- Grader, Thermodynamics, Cornell MAE Department: created rubrics, graded homework, and reported performance to faculty (2024).

CAMPUS INVOLVEMENT

Ultimate Frisbee, Captain & Social Media Chair (2022-26) | Orientation Leader (2022) | Mediocre Melodies (2023-24) | MEng Content Creator (2026-Present)